



One River: Introducing an Integrated Framework and Paradigm for River-Livelihood-Governance Convergence

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Abstract

Rivers are foundational to ecological sustainability, cultural identity, and human well-being. Yet, existing governance models often treat them as fragmented hydrological systems, undermining the intricate interdependencies between ecosystems, livelihoods, and institutions. This paper introduces the One River framework—an integrative paradigm that reconceptualizes rivers as socio-ecological assemblages, advancing a convergence approach to water governance. Building on lessons from *One Health*, *EcoHealth*, and *Planetary Health*, the framework foregrounds four interlinked pillars: ecological integrity, livelihood interdependence, governance integration, and knowledge pluralism. Through critical synthesis and cross-regional case analyses—including the Volta, Amazon, Ganges, Mekong, Mississippi, and Whanganui rivers—the study identifies persistent shortcomings in dominant approaches such as Integrated Water Resources Management and River Basin Organizations, including technocratic bias, institutional fragmentation, and exclusion of Indigenous knowledge systems. The *One River* framework advocates for transdisciplinary collaboration, co-produced knowledge, legal innovations (e.g., river personhood), and adaptive governance attuned to local ecologies and socio-cultural realities. By embedding rivers within networks of relational, upstream–downstream, and human–nature interdependence, this paradigm offers a transformative roadmap for sustainable and just river futures. It calls for a shift in global water governance—from managing rivers as extractive units to stewarding them as living, relational systems central to equitable development and planetary resilience. This study contributes a structured and operationalizable analytical framework for river governance with demonstrated applicability across diverse socio-ecological contexts.

Keywords River governance · Socio-ecological systems · Convergence frameworks · Integrated water resources management · Ecological integrity · Indigenous knowledge · Participatory governance · River rights

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1 Introduction

Rivers are among the most vital yet vulnerable socio-ecological systems on the planet. They sustain biodiversity, support agriculture and fisheries, provide drinking water, generate energy, and shape cultural and spiritual identities. Despite this centrality, river governance remains highly fragmented across sectors, disciplines, and political boundaries. In many regions, particularly in the Global South, responsibilities are divided among environmental ministries, water agencies, agricultural authorities, and local governments, often with overlapping or conflicting mandates (Mehta et al. 2019a, b; Grafton et al. 2020). This fragmentation leads to inefficiencies, policy inconsistencies, and the marginalization of key stakeholders, particularly Indigenous peoples and smallholder communities.

Fragmentation in river governance arises from administrative divisions and jurisdictional overlaps that produce conflicting management approaches and weaken coordination. Empirical evidence shows that even where collaborative arrangements exist, such as in transboundary systems like the Mekong, governance effectiveness remains uneven due to institutional and political constraints (Wu and Matsumoto 2019). At the same time, the application of ecosystem service frameworks in land-use planning remains limited despite extensive theoretical development (Bai et al. 2018; Förster et al. 2015). A persistent disconnect also exists between ecological science, policy frameworks, and lived experience. Scientific approaches often prioritize hydrological metrics while underrepresenting social and cultural dimensions (Boelens 2015), and policy instruments tend to treat rivers as resources to be managed rather than as integrated socio-ecological systems. Consequently, local and Indigenous knowledge systems are frequently excluded from decision-making processes (Krchnak et al. 2011).

In response, sustainability science has increasingly emphasized systems-based and integrative approaches. Frameworks such as “One Health” promote the integration of human, animal, and environmental health through cross-sectoral collaboration (Destoumieux-Garzón et al. 2018), while “Planetary Health” links human well-being to the condition of Earth’s natural systems (Whitmee et al. 2015). Similarly, Nature-Based Solutions (NbS) seek to align ecological restoration with societal benefits (Cohen-Shacham et al. 2016). These approaches highlight the importance of integration but do not fully capture the complexity of river systems, which operate across hydrological, ecological, and governance boundaries and are deeply embedded in place-based livelihoods.

Empirical and interdisciplinary studies further demonstrate that incorporating stakeholder perspectives and participatory governance can enhance policy effectiveness and system resilience (Posner et al. 2016; Shackleton et al. 2019). Nature-based interventions, such as the Netherlands’ “Room for the River” program, illustrate how integrating ecological processes into governance can improve flood management and ecosystem restoration (Oppenheimer et al. 2014; Gleeson et al. 2019). However, challenges remain in ensuring long-term, equitable, and context-specific implementation of such approaches.

This study introduces the “One River” framework to address these gaps. Rather than positioning it as a replacement for existing approaches, the framework is developed as an analytical extension that integrates ecological integrity, livelihood interdependence, governance integration, and knowledge systems into a unified model. It advances the literature by operationalizing these dimensions into measurable components and establishing a causal logic linking governance processes to socio-ecological outcomes. Through comparative

cross-basin analysis, the study demonstrates limitations in existing models and positions the One River framework as a complementary, policy-relevant approach for improving river governance.

2 Methodology

This study adopts a structured scoping review approach to ensure methodological transparency and analytical rigor in developing the One River framework. The approach is informed by socio-ecological systems and integrative research frameworks (Bixler et al. 2019; Bouamrane et al. 2016). Relevant literature was identified through targeted searches in Web of Science, Scopus, and Google Scholar using combinations of keywords such as “river governance,” “integrated water resources management,” “river basin management,” “socio-hydrology,” and “adaptive water governance.”

Searches were limited to peer-reviewed publications in English from 2000 to 2025. The initial search yielded approximately 1,000–1,200 records, which were screened through a three-stage process: title screening, abstract screening, and full-text review. After removing duplicates and irrelevant studies, approximately 180 studies were retained for the final synthesis. Studies were included if they addressed river governance, socio-ecological interactions, or integrative water management frameworks, and excluded if they focused solely on technical hydrology without governance or socio-ecological relevance. A thematic coding approach was applied to identify key dimensions—ecological integrity, livelihood interdependence, governance integration, and knowledge systems—consistent with integrative and participatory knowledge frameworks (Buytaert et al. 2014). These themes were systematically synthesized to construct the analytical structure of the One River framework. A comparative synthesis across selected river basins was conducted to evaluate the applicability and limitations of existing governance approaches.

3 Analytical Structure of the One River Framework

To move beyond a descriptive paradigm, the One River framework is operationalized as an analytical governance model defined by explicit units of analysis, measurable variables, and causal relationships linking governance processes to socio-ecological outcomes.

3.1 Units of Analysis

The framework operates across three interconnected levels. First, river basin systems capture biophysical and ecological processes, including hydrology, biodiversity, and ecosystem dynamics. Second, governance institutions encompass formal and informal regulatory arrangements, policies, and coordination mechanisms across scales. Third, stakeholder systems include communities, resource users, and knowledge holders whose interactions shape river use and management. These units are analytically linked and jointly determine system performance.

3.2 Core Analytical Variables

The framework is structured around four core dimensions that can be empirically assessed. Ecological integrity refers to the condition and functioning of river ecosystems, including water quality, flow regimes, and biodiversity status. Livelihood interdependence captures the extent to which human well-being and economic activities depend on river systems. Governance integration reflects the degree of coordination across sectors, institutions, and spatial scales. Knowledge pluralism refers to the inclusion and integration of scientific, local, and Indigenous knowledge systems into decision-making processes.

3.3 Outcome Variables and Performance Indicators

These dimensions collectively shape key governance outcomes. Ecological sustainability is reflected in reduced degradation and improved ecosystem health. Social equity and inclusion capture the extent of fair participation and benefit distribution among stakeholders. Adaptive capacity and resilience refer to a system's ability to respond to environmental and socio-economic change. Conflict reduction indicates improved coordination and reduced competition over shared resources. Together, these outcomes provide measurable indicators for evaluating framework performance.

3.4 Governance Conditions and Applicability

The framework is most applicable in contexts characterized by institutional fragmentation, strong livelihood dependence on river systems, and complex socio-ecological feedbacks. Under such conditions, integrated and participatory governance approaches are expected to outperform fragmented models such as IWRM.

3.5 Causal Logic of the Framework

The One River framework posits that increased governance integration reduces fragmentation and enhances coordination, while knowledge pluralism improves decision-making legitimacy and contextual relevance. When aligned with livelihood–ecology interactions, these processes reduce resource conflicts and enhance system stability. Together, these mechanisms generate reinforcing feedback that improves ecological integrity, social outcomes, and overall system resilience.

This establishes a clear causal pathway linking governance processes to socio-ecological outcomes, positioning the One River framework as a testable and operational model for river governance.

4 'One River' as a Socio-Ecological Convergence Framework for River Management

The One River framework is structured around three core dimensions—livelihood interdependence, ecological integrity, and governance integration (Fig. 1). These dimensions jointly shape the socio-ecological dynamics of river systems and provide a basis for analyz-

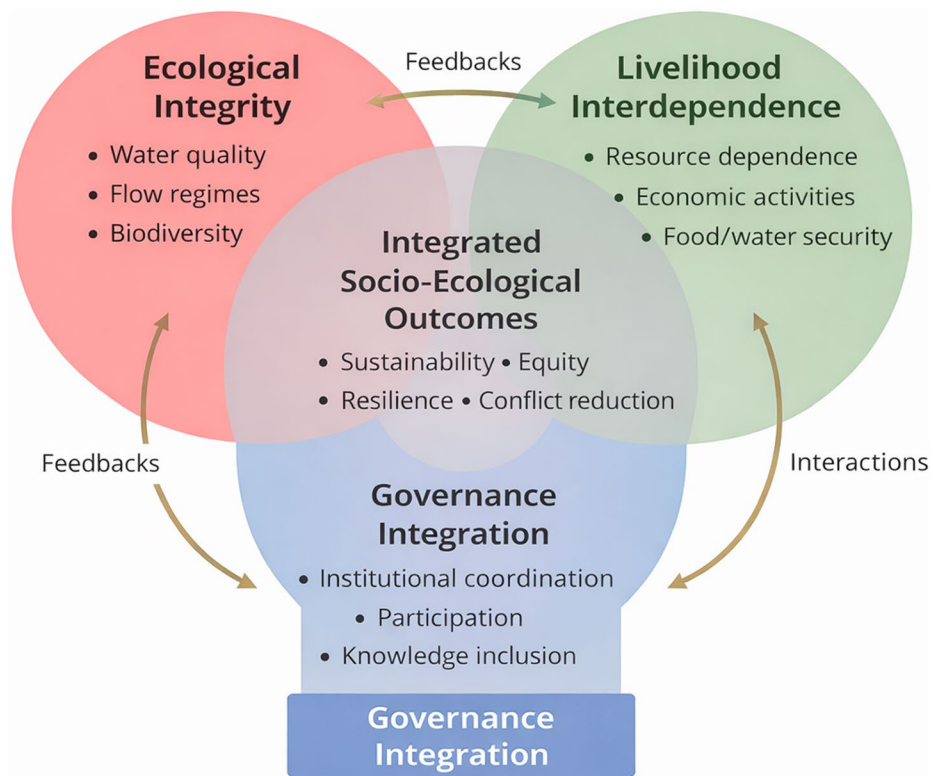


Fig. 1 Analytical representation of the one river framework linking governance, livelihoods, and ecological integrity

ing how ecological processes, human well-being, and institutional arrangements interact. Their intersection represents integrated socio-ecological outcomes, including sustainability, equity, resilience, and conflict reduction. The framework emphasizes coordinated governance, cross-scale interactions, and knowledge integration as key mechanisms for improving river system performance under complex environmental and socio-political conditions.

4.1 Livelihood as a Core Dimension

Livelihoods are closely linked to river systems, which provide essential resources for agriculture, fisheries, and tourism. In regions such as the Mekong River Basin and the Min River, communities adopt diverse livelihood strategies to manage environmental risks and maintain food security. The Sustainable Livelihood Framework (SLF) highlights the role of natural, human, financial, and social capital in shaping resilience under varying governance conditions. For instance, Wang et al. (2019) show that livelihood diversification in mountainous regions reflects adaptive responses to socio-economic pressures and environmental policy shifts.

Livelihoods dependent on river resources are highly sensitive to changes in river health, reinforcing the need for governance approaches that integrate socio-economic and ecologi-

cal objectives (Sithirith et al. 2024; Zhao et al. 2020). This interdependence underscores that effective river governance must simultaneously address livelihood sustainability and ecological integrity.

4.2 Rivers as Integrated Socio-Ecological Systems

Rivers function as interconnected socio-ecological systems that support ecological processes, economic activities, and cultural identities. They underpin livelihoods such as fishing, agriculture, and tourism, while their ecological condition directly influences socio-economic outcomes (Shang et al. 2021). Empirical evidence demonstrates that governance interventions in riverine systems can shape both ecological health and livelihood resilience, as observed in studies of the Meghna Estuary (Talukdar et al. 2022). Institutional arrangements and infrastructure decisions further influence the adaptive capacity of river-dependent communities, highlighting the importance of governance in mediating socio-ecological outcomes (Pal et al. 2023). However, conventional river management has largely emphasized engineering and hydrological control, often overlooking these broader interactions.

In contrast, the One River framework conceptualizes rivers as integrated systems comprising multiple, interdependent dimensions:

- Hydrological systems: flow regimes, sediment transport, and recharge processes.
- Ecological systems: biodiversity, wetlands, and ecosystem functions.
- Economic systems: fisheries, agriculture, and related livelihoods.
- Cultural systems: values, identities, and Indigenous knowledge systems.
- Governance systems: formal, informal, and customary institutions.

This integrated perspective aligns with socio-ecological systems theory, which emphasizes the management of coupled human–natural systems as interconnected and interdependent (Folke et al. 2016).

4.3 Governance as a Bridging Framework

River governance provides the institutional mechanism for operationalizing the One River framework by linking ecological processes with social institutions, legal systems, and participatory practices. Conventional governance approaches remain fragmented, often dominated by sectoral priorities such as agriculture, energy, and urban development, resulting in conflicting mandates and ecological degradation (Schmeier 2013; Mehta et al. 2019a, b). In response, integrated river basin management approaches emphasize coordination across scales and stakeholders to address interconnected challenges, including pollution, land use, and water equity (Hooper 2005).

Polycentric and co-management governance models further enhance adaptive capacity, accountability, and institutional flexibility (Ostrom 2010). Emerging legal innovations, such as the recognition of rivers as legal entities, exemplified by the Whanganui River, reflect a shift toward relational and rights-based governance aligned with Indigenous perspectives (O'Donnell and Talbot-Jones 2018). These developments position governance as the central mechanism for integrating ecological, institutional, and cultural dimensions within river systems.

Governance also mediates interactions between river systems and dependent livelihoods by shaping access to resources, resource allocation, and conflict resolution. Context-specific and participatory governance arrangements improve resilience to environmental change and support equitable outcomes (Zenebe et al. 2021). Aligning governance systems with cultural practices further strengthens stewardship and reinforces socio-ecological sustainability (Mensah and Bukari 2024).

Rather than replacing IWRM, the One River framework is positioned as an analytical extension that addresses persistent implementation limitations. While IWRM has advanced coordinated water management, its application has been constrained by institutional fragmentation, power asymmetries, and limited integration of local knowledge. The One River framework builds on these foundations by explicitly integrating socio-ecological variables, governance dynamics, and knowledge pluralism into a unified analytical model.

4.4 Relational Linkages in River Systems

A central feature of the One River framework is its emphasis on relational dynamics across spatial and socio-ecological scales. Upstream–downstream linkages highlight how activities such as deforestation and mining generate downstream impacts on water quality, flood risk, and biodiversity, requiring coordinated basin-wide governance. Land–water interactions demonstrate how land-use practices influence hydrological processes through erosion, runoff, and pollution, underscoring the need for integrated management approaches. Human–nature relationships emphasize that rivers are embedded within socio-ecological systems, shaping livelihoods, cultural identities, and environmental outcomes, and requiring governance approaches that move beyond purely control-oriented models (Latulippe and Klenk 2020).

These relational linkages support convergence-based governance by integrating environmental, social, and institutional dimensions into a unified analytical perspective. By embedding rivers within multi-scalar and multi-actor systems, the One River framework enables governance approaches that are ecologically informed, socially inclusive, and institutionally coordinated. This strengthens the capacity for co-management, participatory decision-making, and adaptive governance in complex river systems.

5 Limitations of Existing River Management Models

Contemporary river governance models, particularly Integrated Water Resources Management (IWRM) and River Basin Organizations (RBOs), aim to balance social, economic, and environmental objectives through coordinated planning and stakeholder engagement (GWP 2000; Schmeier 2013). However, their implementation has often been constrained by institutional fragmentation, power asymmetries, and limited integration of local knowledge systems (Mehta et al. 2019a, b).

While IWRM promotes coordinated management, it often struggles to be operationalized across complex governance contexts. Similarly, RBOs—such as the Mekong River Commission and the Volta Basin Authority—often prioritize infrastructure and economic objectives while insufficiently addressing equity, ecological integrity, and community participation (Molle 2009). These models also tend to overlook the epistemic and cultural

dimensions of river systems, including Indigenous knowledge and relational values (Boelens 2015). These limitations highlight persistent governance gaps, including fragmented institutional arrangements, technocratic decision-making, weak integration of knowledge, and continued socio-ecological degradation.

5.1 Fragmented Institutions and Policy Silos

River governance is often characterized by fragmented institutional structures and overlapping mandates, hindering coordination across sectors and scales. Conflicts between upstream and downstream actors, as well as among sectors such as agriculture, energy, and conservation, further complicate integrated management (Schipper and Barbarossa 2022).

This fragmentation limits basin-wide coordination and reduces the effectiveness of water management strategies (Yalew et al. 2024). It also contributes to inefficient resource allocation and environmental degradation. For example, prioritization of hydropower development over irrigation needs can intensify socio-economic tensions and negatively affect dependent communities. Addressing these institutional silos requires improved coordination mechanisms and inclusive stakeholder engagement to support more coherent and adaptive governance systems.

5.2 Technocratic and Top-Down Governance

Existing governance models are often shaped by technocratic and top-down approaches that prioritize engineering solutions and economic outputs over socio-ecological considerations (Yalew et al. 2024). This orientation frequently marginalizes Indigenous knowledge systems and local stakeholders, limiting the inclusiveness and effectiveness of decision-making processes (Poricha and Dasgupta 2011). Many governance structures also reflect historical legacies of centralized and extractive water management, reinforcing inequalities and constraining adaptive responses (Forni et al. 2021). As a result, opportunities for integrating local knowledge and community-based solutions remain underutilized. Incorporating participatory governance approaches and diverse knowledge systems is therefore essential for improving equity, legitimacy, and long-term sustainability in river governance (Cheng et al. 2022).

5.3 Inadequate Integration of Knowledge Systems

A key limitation of existing river governance models is the insufficient integration of diverse knowledge systems. Dominant reliance on Western scientific approaches often marginalizes Indigenous and experiential knowledge, resulting in incomplete understanding of river dynamics and socio-ecological interactions (Barnhart et al. 2018). The emphasis on quantitative data alone further constrains comprehensive assessments by excluding context-specific and qualitative insights (Holysh et al. 2018).

Addressing this gap requires co-production of knowledge, in which local stakeholders, scientists, and policymakers collaborate to inform decision-making. Such integration enhances adaptive capacity and enables governance systems to respond more effectively to ecological variability and community needs (Klug et al. 2017).

5.4 Ecological Degradation and Displacement

Governance fragmentation and inadequate integration mechanisms contribute directly to ecological degradation and the displacement of river-dependent communities. Changes in flow regimes, biodiversity loss, and pollution undermine ecosystem resilience and generate significant socio-economic impacts (Esser et al. 2019). Infrastructure development and large-scale projects often exacerbate these effects by altering habitats and reducing resource availability, particularly for fisheries-based livelihoods (López and Haines 2017).

These impacts are disproportionately borne by marginalized communities, which frequently lack secure rights to water and land resources (Rodríguez 2024). Displacement not only disrupts livelihoods but also erodes cultural and social systems, intensifying challenges related to equity and environmental justice (Lorance et al. 2011).

6 Examples of River–Livelihood–Governance Integration

The One River framework provides an integrative approach to river governance by linking ecological integrity, livelihood interdependence, and institutional coordination (Fig. 1). The following cases illustrate how variations in governance integration and knowledge inclusion shape socio-ecological outcomes across river basins. Comparative analysis indicates that systems with lower levels of coordination and knowledge pluralism tend to exhibit weaker ecological and social performance, underscoring the analytical relevance of the One River framework.

6.1 Volta River Basin (West Africa)

The Volta Basin illustrates tensions between hydropower development and livelihood sustainability. Large dams, including Akosombo and Bui, have altered ecological flows and disrupted fisheries and agriculture (Bandaragoda and Babel 2010; Graham et al. 2016). While the Volta Basin Authority promotes transboundary coordination (VBA 2021), governance challenges persist due to limited stakeholder inclusion and weak integration of local knowledge (Teye and Yaro 2017). Applying the One River framework emphasizes participatory governance, ecological flow management, and livelihood-sensitive planning to improve equity and ecological outcomes (Paredes-Arquiola et al. 2014; Anderson et al. 2019; Evers 2016).

6.2 Amazon Basin

The Amazon Basin highlights conflicts between extractive activities and Indigenous governance systems. Despite formal recognition of Indigenous territories, enforcement remains weak, allowing continued ecological degradation (Acuña et al. 2020). Indigenous knowledge systems play a critical role in sustaining ecological integrity but are often excluded from governance processes (Nustad et al. 2018; Drouineau et al. 2014). The One River framework supports polycentric governance that integrates Indigenous knowledge with formal institutions, enhancing ecological resilience and reducing conflict (Anderson et al. 2019).

6.3 Ganges River (India/Bangladesh)

The Ganges Basin reflects the challenges of pollution, institutional fragmentation, and cultural misalignment. Despite major restoration initiatives, governance remains fragmented across multiple agencies, limiting effectiveness (Erwin et al. 2011; Das and Dasgupta 2011). Pollution pressures continue to undermine ecological and public health outcomes (Grantham et al. 2013). The One River approach highlights the need to integrate cultural values and community practices into governance systems to improve participation, reduce conflict, and enhance restoration outcomes (Jordà-Capdevila et al. 2021; Lokgariwar et al. 2013).

6.4 Mekong River Basin

The Mekong Basin demonstrates transboundary governance challenges driven by hydro-power expansion and uneven institutional authority. Large-scale dam construction has disrupted flow regimes, fisheries, and sediment dynamics (Grumbine and Xu 2011; Wantzen 2023). While institutions such as the Mekong River Commission facilitate coordination, enforcement remains limited (DeLong et al. 2021; Schmeier 2013). The One River framework emphasizes basin-wide coordination, livelihood integration, and participatory governance to strengthen resilience and reduce transboundary tensions (Pharino et al. 2016; Ostrom 2010; Lokgariwar et al. 2013).

6.5 Mississippi River Basin (USA)

The Mississippi Basin faces challenges related to agricultural runoff, fragmented governance, and ecological degradation. Nutrient pollution has led to hypoxia in the Gulf of Mexico, reflecting systemic failures in watershed management (Turner and Rabalais 2003; Fliervoet et al. 2013). Governance remains fragmented across federal and state institutions, with limited integration of Indigenous perspectives (Cosens et al. 2018; Zuo and Liang 2015). The One River framework supports watershed-scale coordination, inclusion of diverse stakeholders, and integration of ecological and agricultural systems to improve outcomes (Kang et al. 2024; Grantham et al. 2019; Hassan and Ansari 2010).

6.6 Okavango River Basin

The Okavango Basin demonstrates relatively strong transboundary cooperation through the Permanent Okavango River Basin Water Commission (OKACOM), which emphasizes sustainability and data sharing (OKACOM 2021). Community-based natural resource management approaches further support local livelihoods and ecosystem conservation (Roe et al. 2009). This case reflects alignment with One River principles, particularly in balancing ecological integrity, livelihoods, and governance coordination.

6.7 Whanganui River (New Zealand)

The Whanganui River represents a landmark case of legal personhood and Indigenous co-governance. Recognition of the river as a legal entity integrates Māori worldviews into

governance structures, supported by shared institutional arrangements (Charpleix 2018; O'Donnell and Talbot-Jones 2018).

This model exemplifies knowledge pluralism, relational governance, and institutional integration, providing a benchmark for applying the One River framework in practice. To synthesize insights across the selected river basins and enable systematic comparison, Table 1 presents a structured overview of key governance characteristics, identified gaps, and observed socio-ecological outcomes. It highlights how varying governance configurations influence the extent to which integrative dimensions—ecological, livelihood, governance, and knowledge systems—are addressed across contexts.

7 Parallels of One River with Convergence Frameworks

The One River framework aligns with established integrative paradigms such as One Health, EcoHealth, and Planetary Health, all of which emphasize the interconnectedness of ecological systems, human well-being, and governance structures. Table 2 provides a comparative synthesis of these frameworks, highlighting their scope, principles, and sectoral focus. One Health emphasizes linkages between human, animal, and environmental health

Table 1 Comparative analysis of river–livelihood–governance systems across selected river basins

Basin	Key challenge	Governance structure	Missing one river dimensions	Observed outcomes
Volta	Hydro-power vs. livelihoods	Transboundary RBO	Weak participation and low knowledge integration	Livelihood displacement and ecological stress
Amazon	Extractive pressure	National + weak enforcement	Limited governance integration and Indigenous marginalization	Conflict and biodiversity loss
Ganges	Pollution, fragmentation	Multi-level governance	Poor integration, weak cultural alignment	Persistent degradation
Mekong	Trans-boundary dams	Regional commission	Weak enforcement and limited equity	Flow disruption and livelihood risks
Mississippi	Agricultural runoff	Federal–state fragmented	Low integration and weak Indigenous inclusion	Hypoxia and ecosystem decline
Okavango	Climate variability	Trans-boundary cooperation	Moderate integration and limited scaling	Relative resilience
Whanganui	Legal recognition	Co-governance model	Strong integration across dimensions	Improved stewardship and equity

Table 2 Comparative dimensions of some integrative health and environmental frameworks: one river, one health, EcoHealth, and planetary health

Dimension	One river	One health	EcoHealth	Planetary health
Focus	River systems & livelihoods	Human–animal–environment health	Ecosystems & health	Human health & Earth systems
Principles	Integration, equity, co-governance	Collaboration, coordination	Systems thinking, participation	Sustainability, equity
Sectors	Water, energy, agriculture, governance	Public & animal health	Ecology & public health	Health, climate, economy
Scale	Basin to transboundary	Global/local	Regional/community	Global
Governance	Basin-level, participatory	Cross-sector coordination	Community-based	Global policy
Knowledge	Science + Indigenous knowledge	Interdisciplinary science	Strong local integration	Interdisciplinary
Key drivers	Pollution, dams, and climate	Diseases, AMR, pandemics	Land use, biodiversity loss	Climate change, consumption
Monitoring	Basin monitoring systems	Disease surveillance	Community monitoring	Planetary indicators
Engagement	High (multi-stakeholder)	Medium–high	High (community-led)	Medium
Cultural role	Central	Limited	Moderate	Emerging
Resilience	Water–livelihood resilience	Health system resilience	Socio-ecological resilience	System-wide resilience
Added value	Integrates water, livelihoods, governance	Disease prevention	Ecosystem-based health	Global sustainability framing

through interdisciplinary collaboration (Zinsstag et al. 2011), while EcoHealth focuses on ecosystem–society interactions and participatory approaches to environmental management (Mehring et al. 2017). Planetary Health extends this perspective by situating human health within global ecological limits (Steffen et al. 2015). These frameworks share core principles with the One River approach, including systems thinking, cross-sector integration, and adaptive governance.

River systems exemplify these interconnections, as pressures from urbanization, agriculture, and industrial activity simultaneously affect ecological integrity and social vulnerability (Wei et al. 2022). The One River framework extends convergence thinking to river governance by integrating water systems with livelihoods, governance structures, and knowledge systems (Mohtar et al. 2024). Existing governance approaches have often prioritized technical management objectives while overlooking socio-ecological complexity and equity considerations (Gilvear et al. 2012). Integrative frameworks demonstrate that inclusive and participatory approaches improve system resilience and governance effectiveness, particularly when marginalized perspectives are incorporated (Costanza et al. 2014).

Evidence further indicates that fragmented water governance contributes to unsustainable outcomes by failing to account for system interdependencies (Bouamrane et al. 2016). In contrast, convergence-based approaches support coordinated governance, improved environmental monitoring, and adaptive responses to emerging risks (Frieze and Nuyts 2017). The application of cumulative risk perspectives also highlights the importance of addressing multiple, interacting stressors within socio-ecological systems (Williams et al. 2012; Mohtar et al. 2024). The One River framework reinforces these principles by emphasizing governance integration, knowledge pluralism, and socio-ecological resilience. It supports adaptive management strategies that address complex environmental challenges while integrating diverse knowledge systems and stakeholder perspectives (Genaidy et al. 2017; Özdemir 2020).

In addition, the framework aligns with ecological economics by recognizing the interdependence of natural and human capital in sustaining ecosystem services and livelihoods (Zinsstag et al. 2011). Strengthening governance structures to incorporate local and Indigenous knowledge further enhances decision-making and promotes equitable outcomes (Sánchez-Cañizares et al. 2017). Technological innovation also plays a role in advancing integrated governance. Collaborative monitoring systems and data-driven approaches can improve environmental assessment and support proactive management of river systems (Banerjee et al. 2023). The One River framework builds on convergence-based approaches by explicitly integrating river systems into broader socio-ecological governance models. It provides an analytical extension that links ecological integrity, livelihoods, and governance, offering a structured basis for improving sustainability, equity, and resilience in water management (Henderson and Loreau 2021).

8 Operationalizing the One River Framework

Operationalizing the One River framework requires institutional mechanisms that integrate ecological management, livelihood systems, and participatory governance across scales. Implementation depends on coordinated action among policymakers, scientists, local communities, and basin-level institutions, supported by adaptive management approaches responsive to dynamic socio-ecological conditions (Gran et al. 2019; López et al. 2019). The framework emphasizes governance systems that align ecological integrity with social equity and stakeholder inclusion.

8.1 River Observatories and Transdisciplinary Hubs

River observatories and transdisciplinary hubs provide platforms for integrated monitoring, knowledge co-production, and decision-making. These systems combine scientific data with local knowledge to improve environmental assessment and governance outcomes (Bixler et al. 2019; Boelens et al. 2022). Examples such as early warning systems in trans-boundary basins demonstrate how participatory monitoring enhances data reliability and policy relevance (Vargas et al. 2020; Stokral 2021). By aligning research with local needs, these hubs strengthen adaptive governance and support evidence-based decision-making (Mitchell et al. 2017).

8.2 Citizen Science and Community Stewardship

Citizen science initiatives enhance governance by involving communities directly in monitoring and management processes. Programs such as river guardianship and participatory monitoring improve environmental awareness and strengthen local stewardship (Wong 2016; Lazurko et al. 2023). Community participation in data collection and decision-making increases legitimacy and improves the integration of local knowledge into governance systems (McDermott et al. 2013; Gran et al. 2019). These approaches support more inclusive and context-sensitive river management.

8.3 Legal Personhood and Rights-Based Governance

The recognition of rivers as legal entities represents a shift toward rights-based governance. Cases such as the Whanganui River illustrate how legal personhood can integrate ecological protection with cultural values and Indigenous governance systems (Chastin et al. 2016; Yu et al. 2014). This approach strengthens environmental accountability and aligns governance with relational perspectives that prioritize ecosystem integrity and social equity (Eriksson et al. 2014; Ely et al. 2020; Petts et al. 2006).

8.4 Cross-Sectoral Coordination and Legal Pluralism

Effective implementation requires coordination across institutions and the integration of statutory and customary governance systems. Legal pluralism enables the incorporation of Indigenous knowledge and traditional practices into formal governance structures (Boelens et al. 2022; Stokral 2021). Flexible institutional arrangements that support co-management improve coordination, reduce conflict, and enhance system resilience (Petrů and Tomášková 2020; Bixler et al. 2019).

8.5 Financing and Equity Mechanisms

Sustainable implementation depends on financing mechanisms that align ecological restoration with social equity. Approaches such as Payment for Ecosystem Services and climate adaptation funding can support both environmental and livelihood outcomes (Polk and Sapienza 2008; Lazurko et al. 2023). Embedding equity considerations—distributive, procedural, and contextual—into funding strategies improves governance outcomes

and ensures benefits reach vulnerable populations (Wong 2016; Edronova and Маслакова 2019). Linking financial resources to local participation strengthens long-term sustainability (Ely et al. 2020; Petts et al. 2006).

9 Conclusion

As river systems across the globe face escalating pressures from climate change, unsustainable development, and fragmented governance, there is an urgent need for a new integrative approach—one that recognizes rivers not merely as physical or economic resources, but as relational, socio-ecological entities that sustain life and culture. The One River paradigm provides such a framework. Drawing on convergence models such as One Health and Planetary Health, it reimagines river governance through a lens of ecological integrity, livelihood interdependence, governance integration, and knowledge pluralism. By bridging the divides between upstream and downstream, land and water, science and tradition, and institutions and communities, One River offers a roadmap for integrative, decolonial, and ecologically grounded development. It underscores the importance of recognizing rivers as living systems with rights and relationships, rather than fragmented units to be exploited or managed in isolation. Operationalizing this framework will require bold policy shifts, transdisciplinary academic rethinking, and above all, community-led stewardship rooted in justice and resilience. As water crises deepen, adopting the One River approach is not just an intellectual or ethical imperative—it is a practical necessity for building sustainable futures.

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Declaration

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